COMPARATIVE SYNOPTIC CONTROLS OF POLLUTION EXTREMES

Mexico City vs Hong Kong (2012–2024)

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SEE4994 - Final Presentation

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WHY STUDY EXTREME-POLLUTION DAYS?

- Both cities = megacities → dense population + chronic pollution exposure
- Complex topography & meteorology shape extreme events
- Clear need for synoptic understanding (policy + public health)
- For HK: cyclone-modulated & transport-driven
- For CDMX: basin confinement + stagnation

RESEARCH FOCUS

Research Question

What large-scale circulation pattern governs extreme pollution days in Mexico City, and how does it compare with Hong Kong's regional meteorology and health-based indices?

Sub-objectives

- Characterize pollutant patterns (2012–2024)
- Build and refine the composite methodology (500 hPa)
- Compare with HK meteorological regimes (literature)

STUDY WORKFLOW

6-Stage Guided Study Structure

Week 1: Exploratory analysis

Week 2: Full dataset assembly (2012–2024)

Week 3: Seasonality + early literature

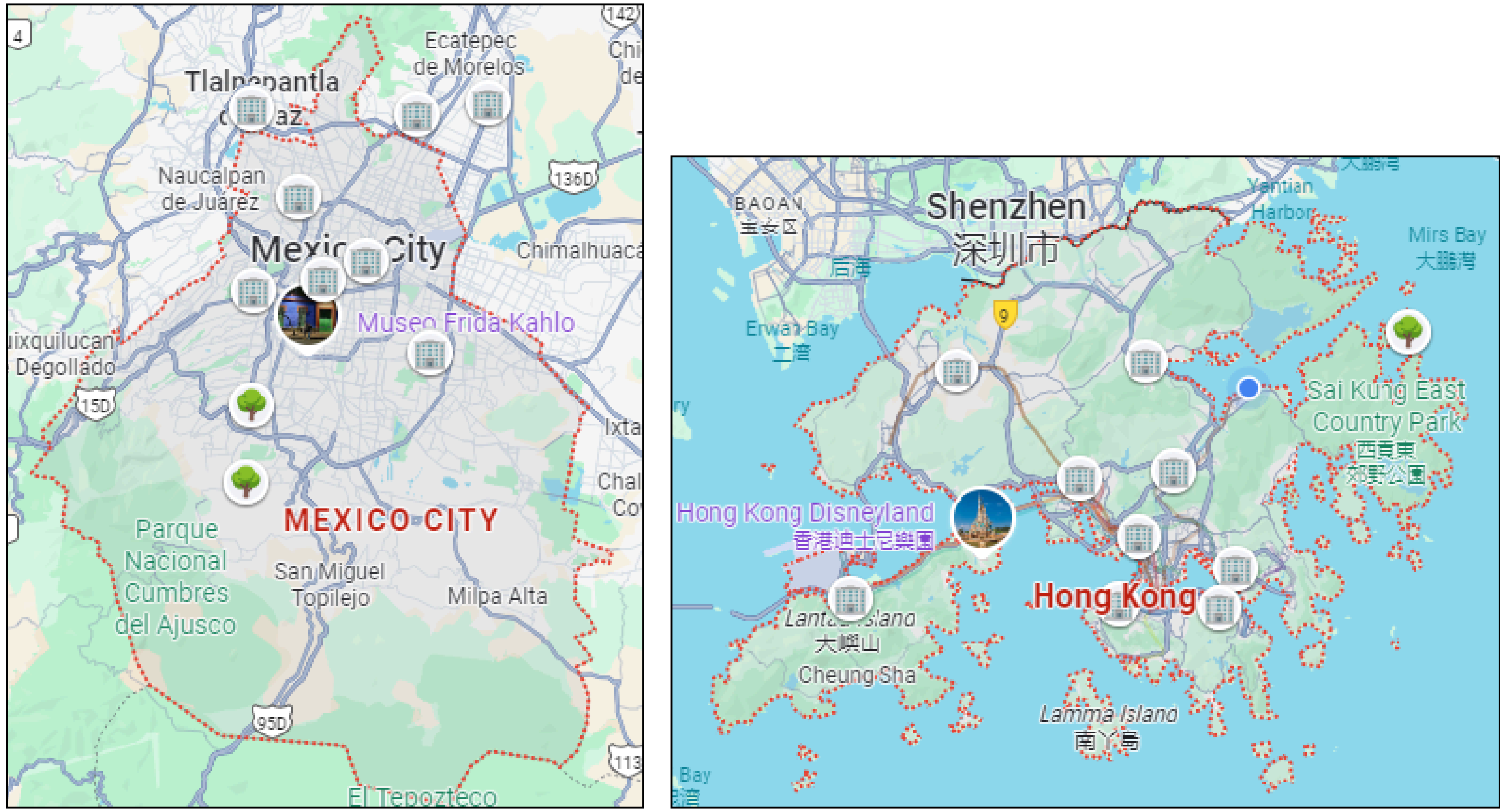
Week 4: Deep MX literature review

Week 5: First NARR composites (500/850 hPa)

Week 6: Revised composites + HK comparison

STUDY REGIONS

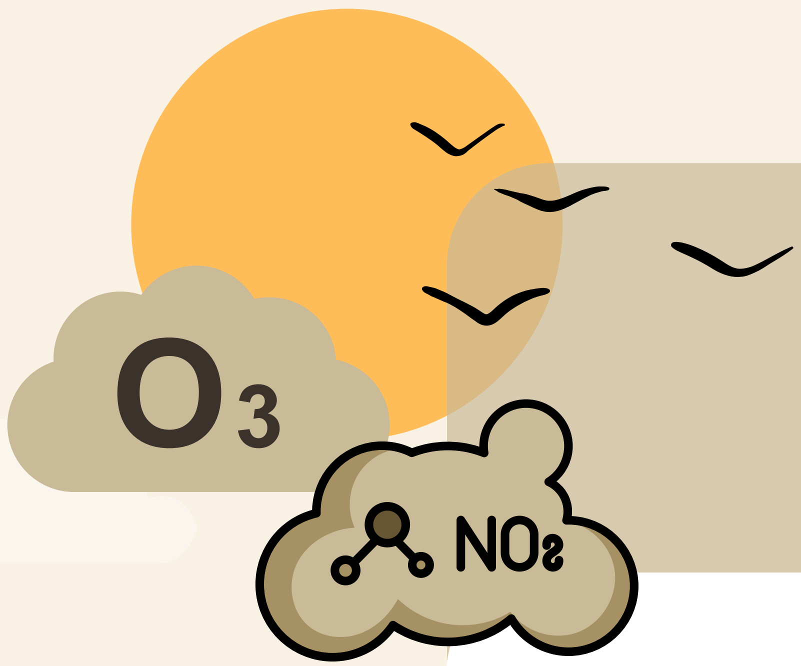
& Monitoring Networks



(a) (b)

Fig. 1. Spatial distribution of the monitoring stations used in this study. (a) MCMA stations; (b) Hong Kong stations. Urban stations (building icon) and rural stations (tree icon) are marked for visual clarity. Station coordinates and metadata were obtained from UPIITA/IPN dataset and AQHI/EPD official sources.

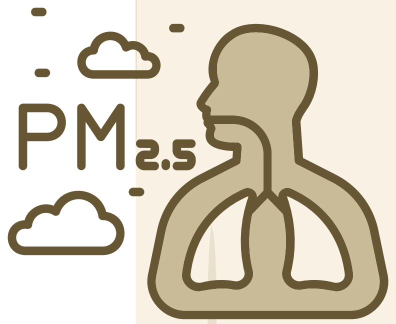
- MCMA: basin at ~2300 masl → strong confinement
- HK: coastal + monsoon + cyclone-modulated
- 10 stations per city, multi-pollutant, 2012–2024



DATASET OVERVIEW

Surface Pollutant Datasets (2012–2024)

- FIve pollutants: $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2
- CDMX → manually downloaded + cleaned (100+ files)
- HK → directly downloadable from EPD
- Harmonized into identical column structure
- Daily means → long-term seasonality + p90 calculation



SEASONALITY

Monthly Mean Concentrations (2012–2024)

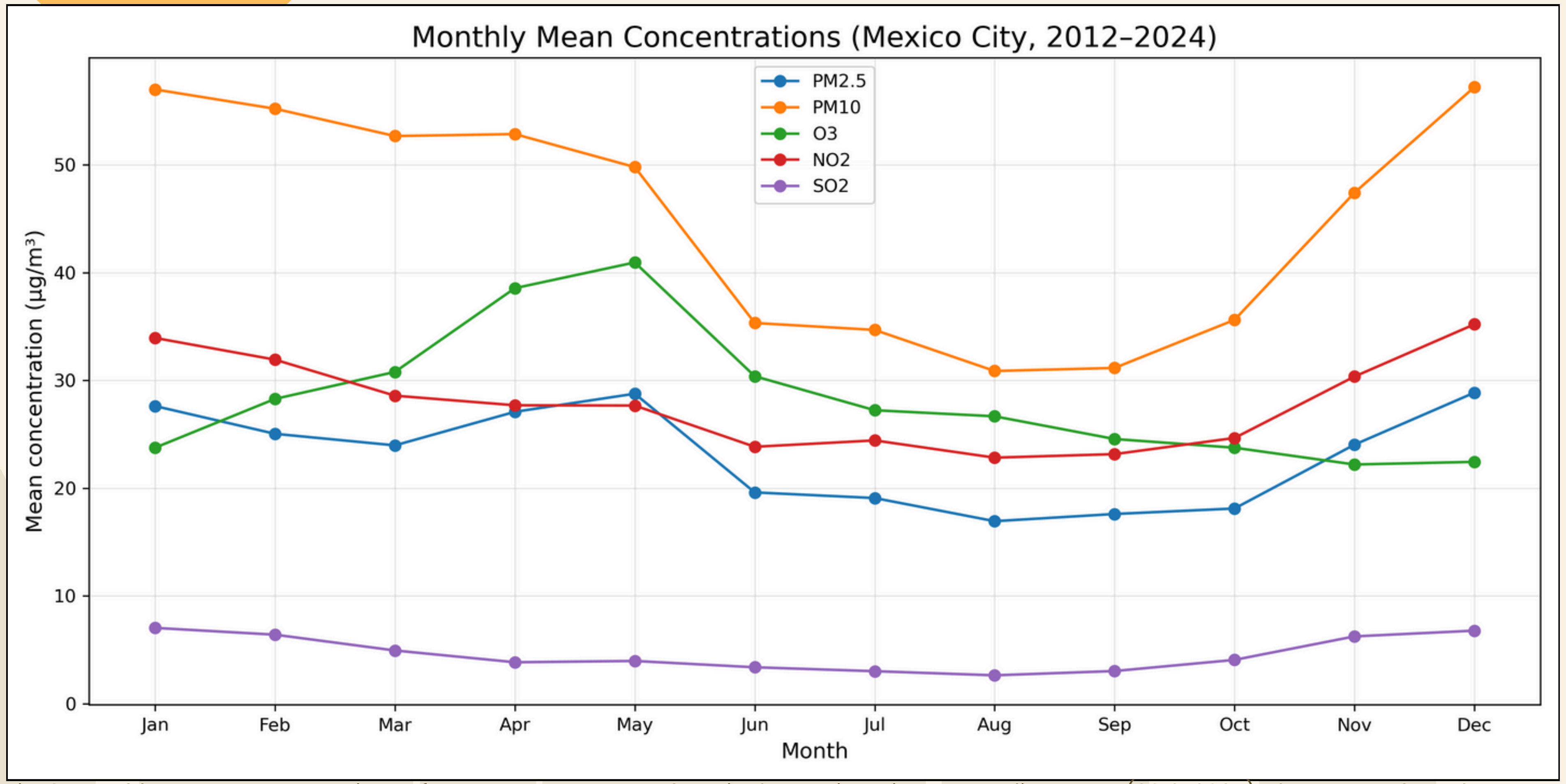


Fig. 2. Monthly mean concentrations of $PM_{2.5}$, PM_{10}, O_3, NO_2 and SO_2 in the Mexico City Metropolitan Area (2012–2024). The curves show the annual cycle of each pollutant based on daily means aggregated by month.

- PM_{10} highest overall
- O_3 peak in April–May
- $PM_{2.5}$ → winter + late-spring rise
- NO_2 → winter + early spring
- SO_2 → low but winter enhancement

DEFINING EVENT DAYS (P90)

Identifying Extreme-Pollution Days (p_{90} method)

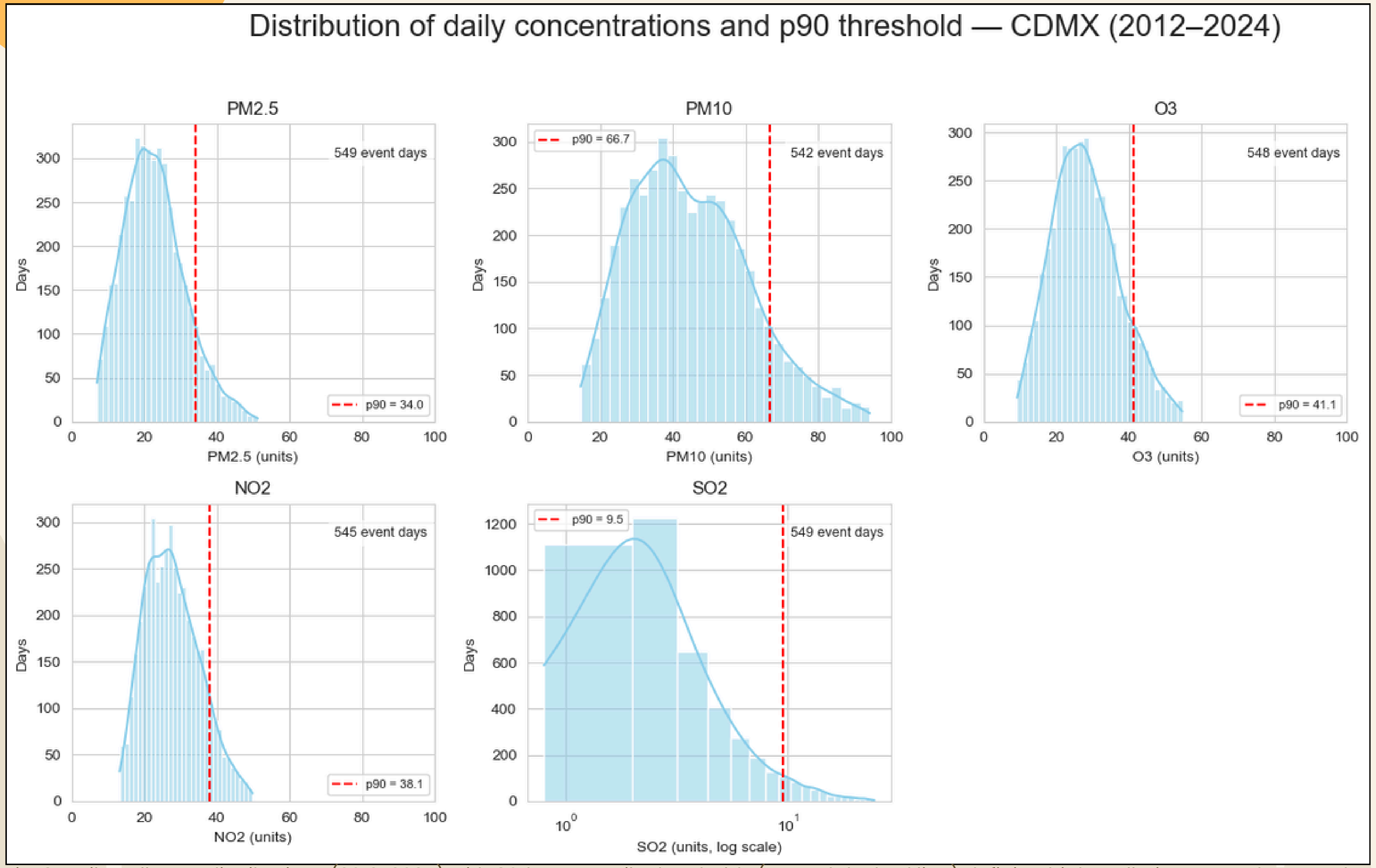


Fig. 3. Daily pollutant distributions (2012–2024) with 90th-percentile thresholds (p_{90}, red dashed line) defining high-pollution event days used in the composite analysis, with uniform and standardized x-axis scale for PM2.5, PM10, O3 & NO2, and logarithmic scale for SO2.

- p_{90} computed per month → captures seasonality
- Top ~10% of days per pollutant
- ~542–549 event days per pollutant
- Used to extract NARR fields for compositing

NARR & COMPOSITE SETUP

500-hPa Composite Framework

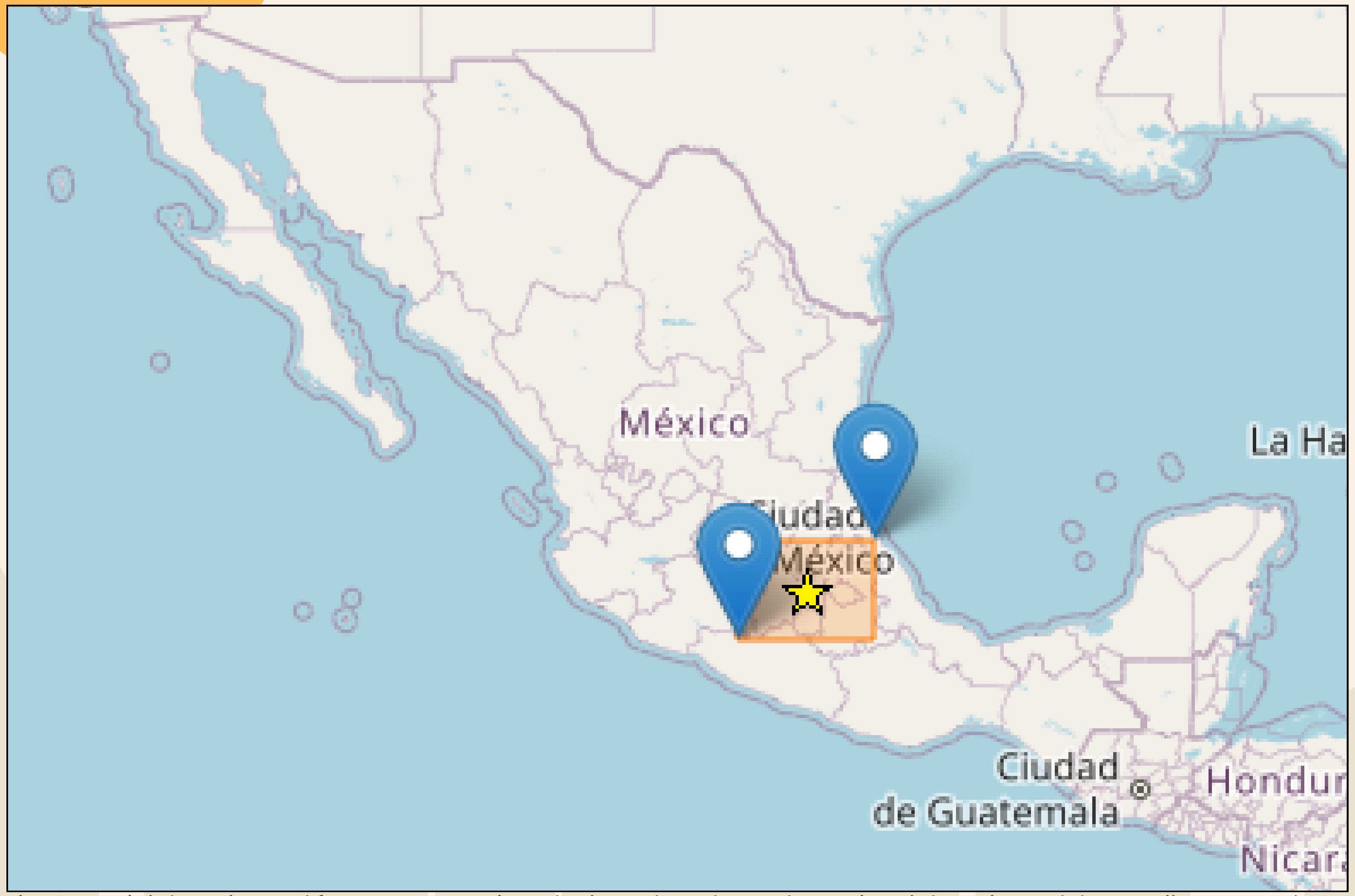


Fig. 4. Spatial domain used for NARR extraction. The large box shows the regional domain, and the smaller orange box outlines the Mexico City Metropolitan Area. The yellow star marks downtown Mexico City.

- NARR reanalysis (NOAA)
- Domain: Mexico + surroundings (12°–33°N, 85°–120°W)
- 31-day smoothed daily climatology (to compute H′)
- Composite = H′ + wind vectors + significance mask
- Basin box + CDMX marker for interpretation

BEFORE/AFTER CORRECTION

Correcting the Anomaly Definition

Week 5:
anomaly = event day
− global mean
(incorrect)

BEFORE

Week 6:
anomaly = event day
− DOY climatology
(correct)

AFTER

- Eliminated inconsistencies between 500/850 hPa
- Enabled coherent synoptic interpretation

500 hPa
SINGLE-POLLUTANT COMPOSITES

- Positive H′ over central Mexico
- Weak easterly / ENE flow (< 2 m/s)
- Basin under ridge/stagnation regime
- Slight differences in spatial extent per pollutant

All pollutants share same synoptic backbone

PM2.5 COMPOSITE

$PM_{2.5}$ – 500 hPa H′ composite

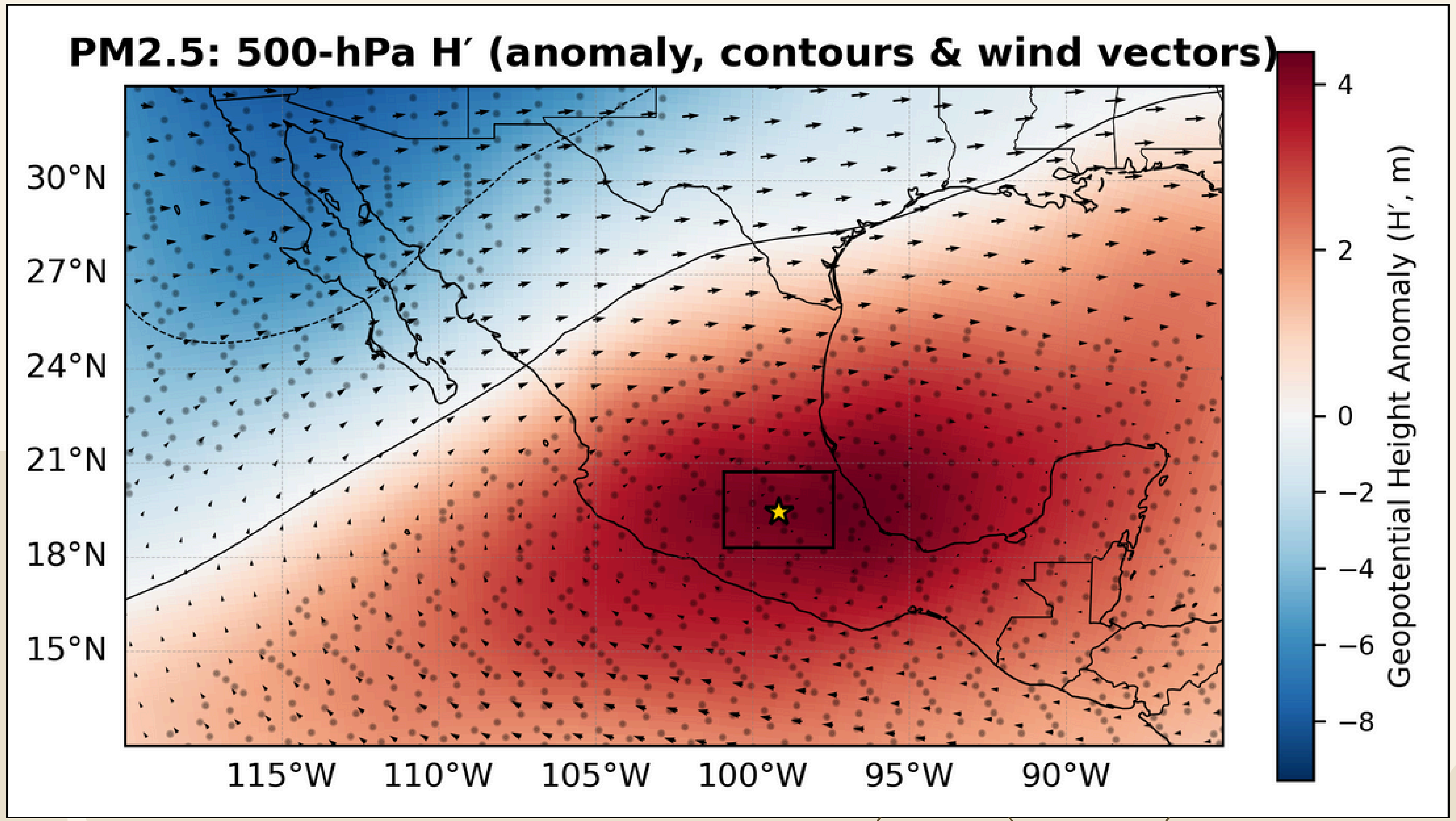


Fig. 5. $PM_{2.5}$ composite at 500 hPa showing geopotential-height anomalies H′ (shading, m), H′ contours (solid = positive; dashed = negative) and mean 500-hPa wind vectors for $PM_{2.5}$ event days (2012-2024). The MCMA box and Mexico City marker are included for reference

- Broad positive H′ centered near MCMA.
- Weak ENE flow (~1.7 m/s) above the basin → low ventilation.
- Consistent with fine-particle build-up under warm, clear, subsident conditions.

Subtropical ridge days strongly favor $PM_{2.5}$ accumulation.

PM10 COMPOSITE

PM_{10} - strongest ridge signal

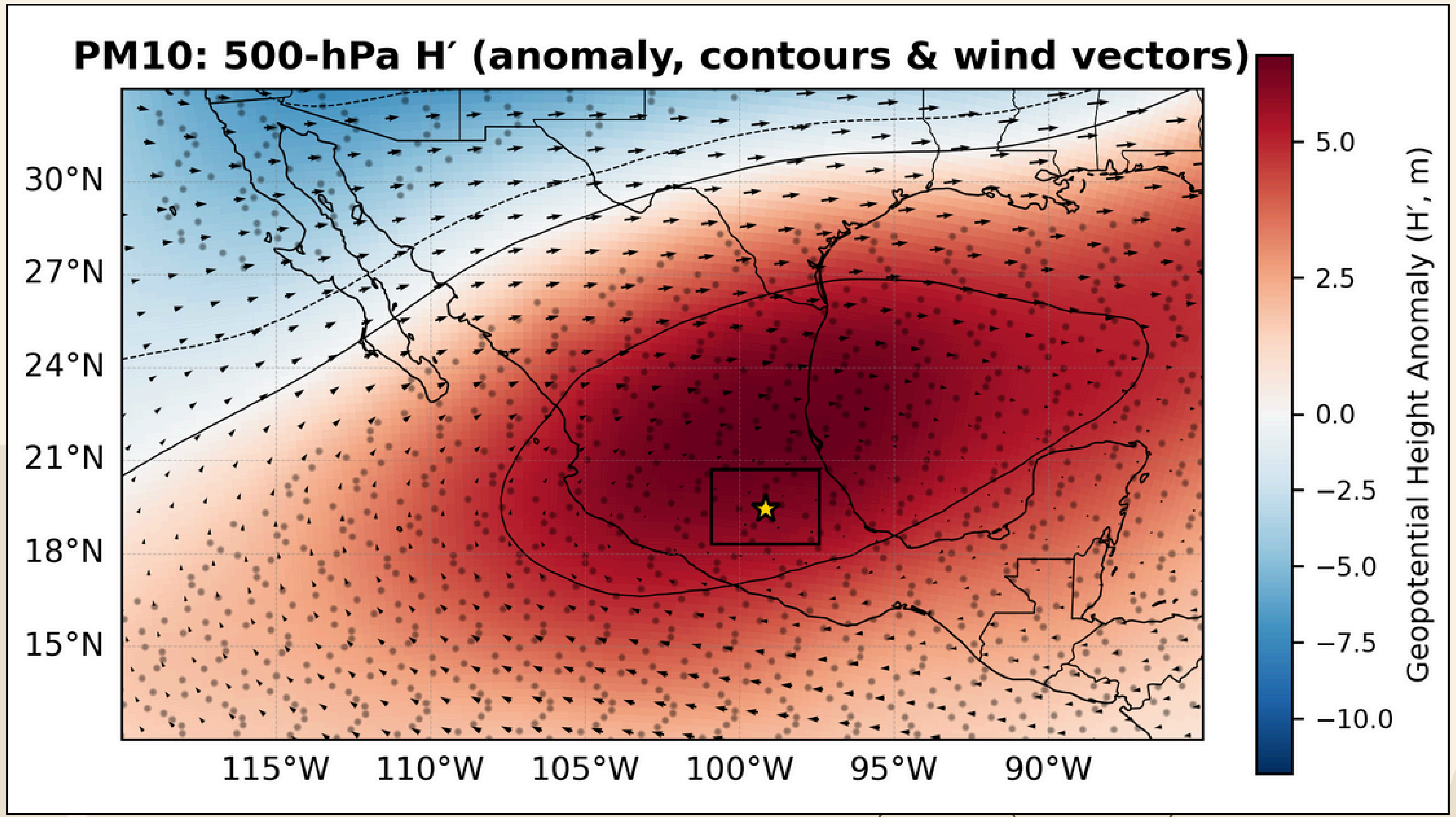


Fig. 5. PM_{10} composite at 500 hPa showing geopotential-height anomalies H′ (shading, m), H′ contours (solid = positive; dashed = negative) and mean 500-hPa wind vectors for PM_{10} event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

- Largest H′ (~+6 m) among pollutants; closed positive contours over MCMA.
- Very weak NE→ENE flow (~0.9 m/s).
- Dry, stagnant episodes favor coarse particles and re-suspension.

PM_{10} peak days are textbook "blocking + stagnation."

O3 COMPOSITE

O_3 - compact ridge and minimal winds

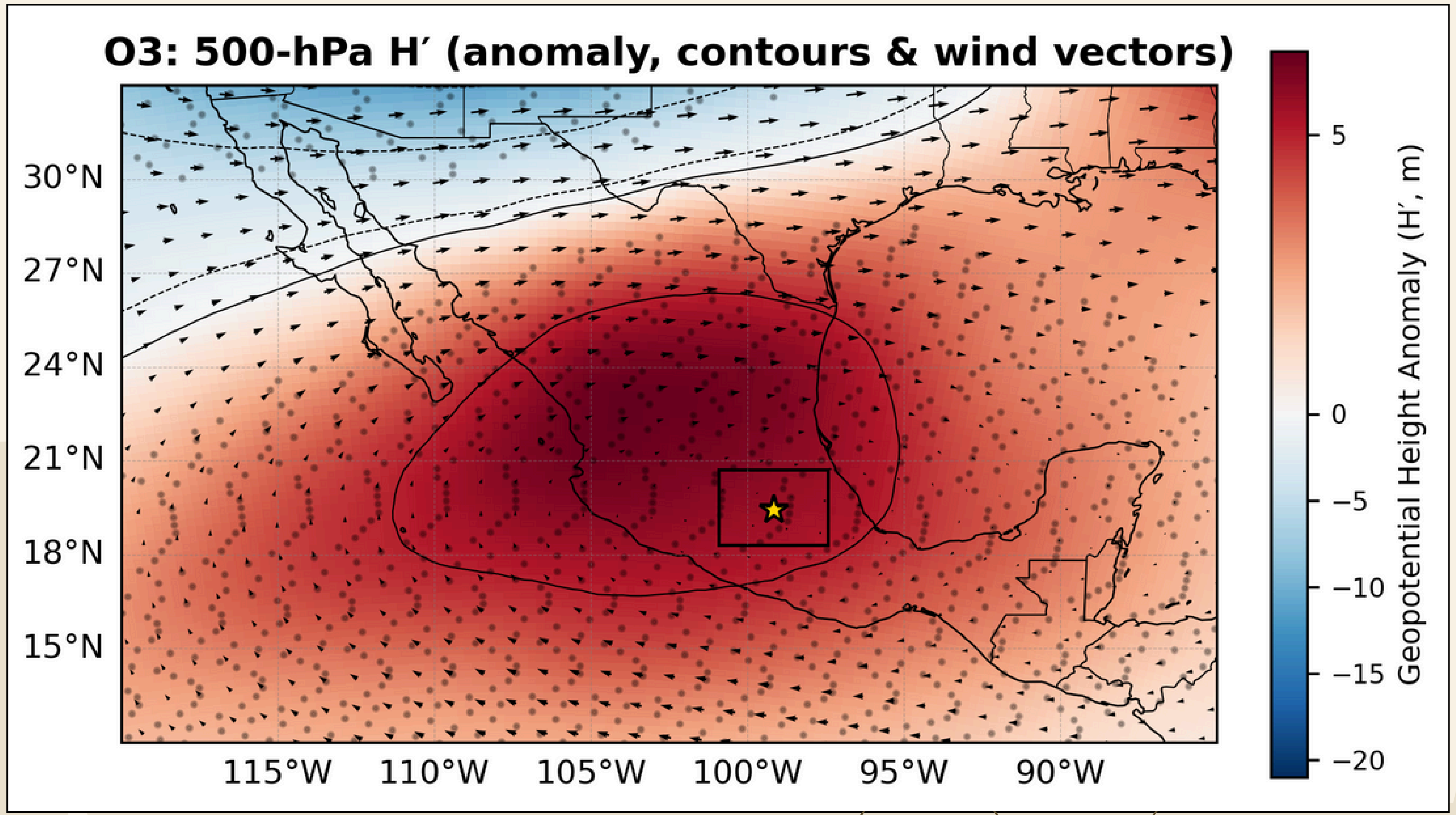


Fig. 6. O_3 composite at 500 hPa showing geopotential-height anomalies H′ (shading, m), H′ contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for O_3 event days (2012–2024). The MCMA box and Mexico City marker are included for reference.

- Compact positive H′ core (~+5.6 m).
- Weakest winds across species (~0.75 m/s).
- Clear skies + stability → enhanced photochemistry, limited dispersion.

O_3 peaks align with warm, subsident, low-wind days.

NO2 COMPOSITE

NO_2 - primary-pollutant stagnation

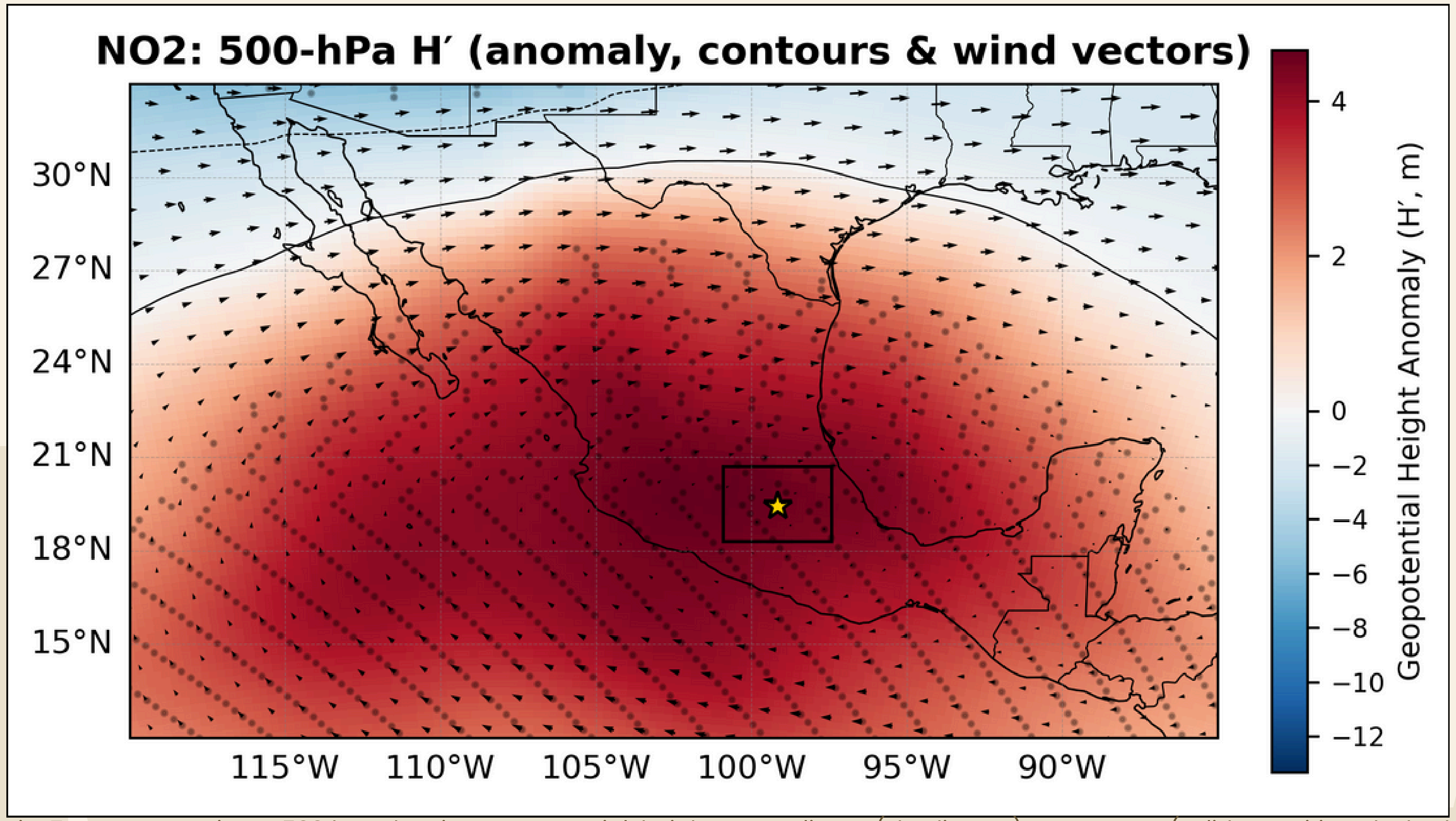


Fig. 7. NO_2 composite at 500 hPa showing geopotential-height anomalies H′ (shading, m), H′ contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for NO_2 event days (2012–2024). The MCMA box and Mexico City marker are included for reference.

- Basin-scale ridge (H′ ~ +4.5 m).
- Light E–NE flow (~1.35 m/s).
- Weak vertical mixing → near-surface accumulation inside the valley.

Anticyclonic stability suppresses NO_2 ventilation.

SO2 COMPOSITE

SO_2 - broader ridge footprint

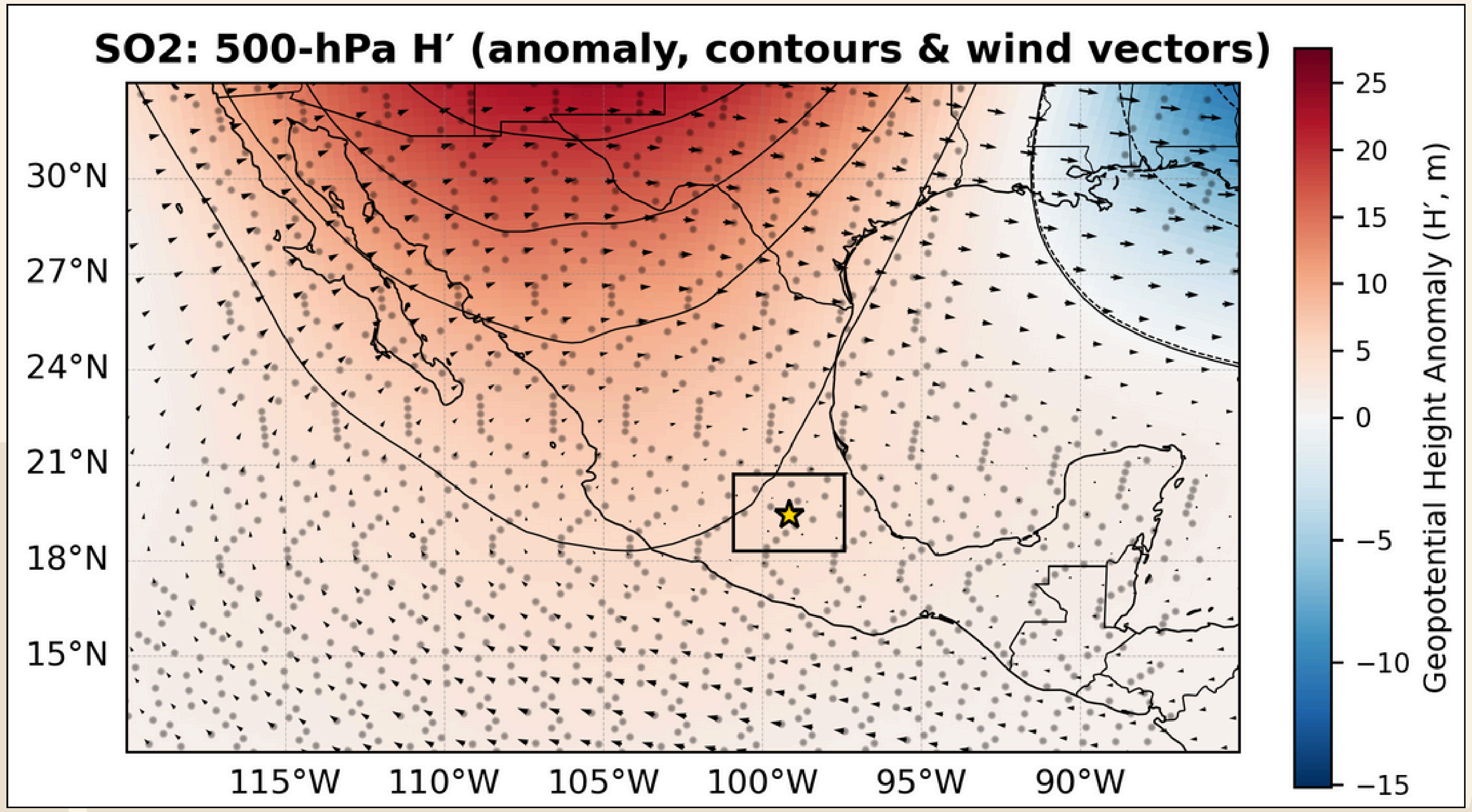


Fig. 8. SO_2 composite at 500 hPa showing geopotential-height anomalies H′ (shading, m), H′ contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for SO_2 event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

- Positive H′ over MCMA with ridge limb extending NW.
- Very weak E–NE flow (~0.8 m s^{-1}).
- Suggests regional-scale subsidence › tight local core.

Same mechanism, broader spatial control.

MULTIPANEL COMPARISON

Multipollutant Synoptic Fingerprint

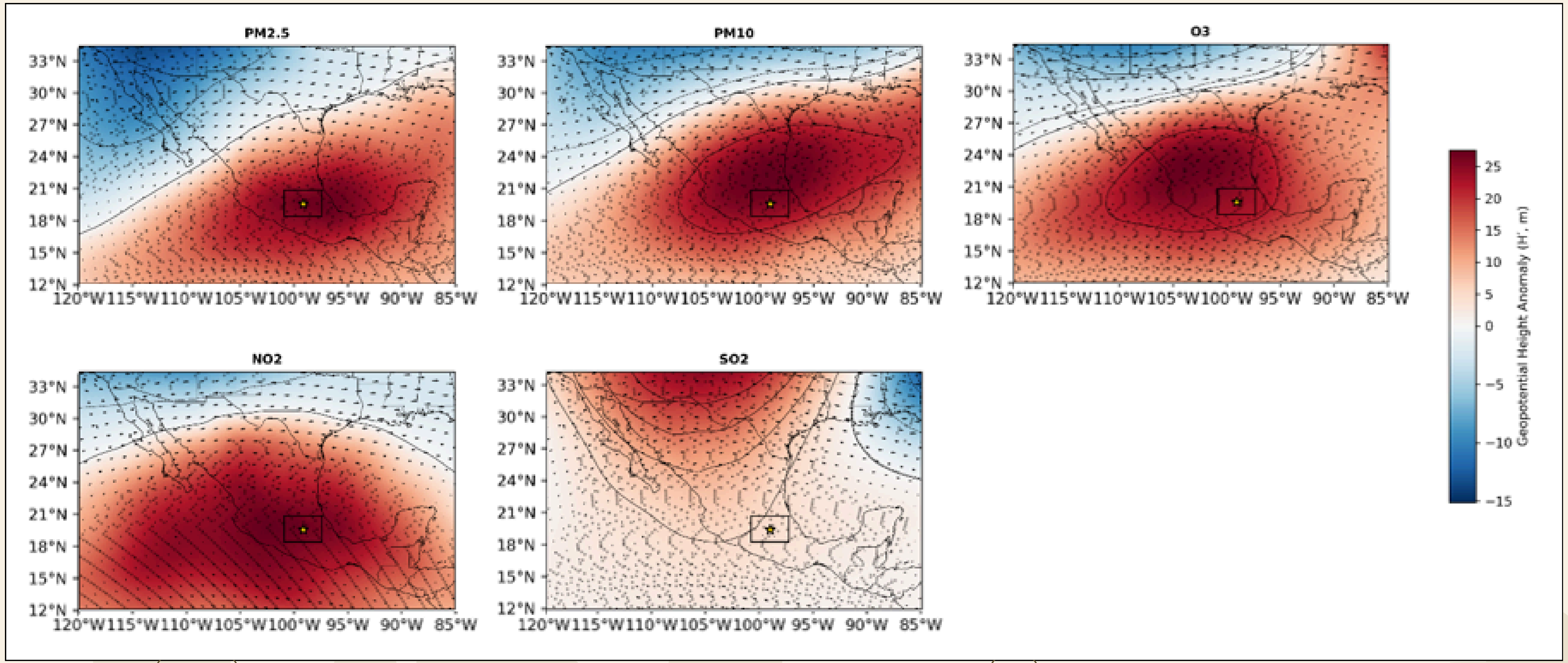


Fig. 9. Multipanel (500 hPa): Composites for $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2 showing a coherent ridge-stagnation pattern ($H' > 0$) and weak ENE flow during high-pollution days. Stippling marks $p < 0.05$. (Created by the author).

- All pollutants → ridge signature
- H′ ~ +4 to +6 m
- Weak easterlies over MCMA
- Only minor differences in amplitude/localization

Unified meteorological driver.

INTERPRETATION OF MECHANISM

Ridge–Stagnation Mechanism Over MCMA

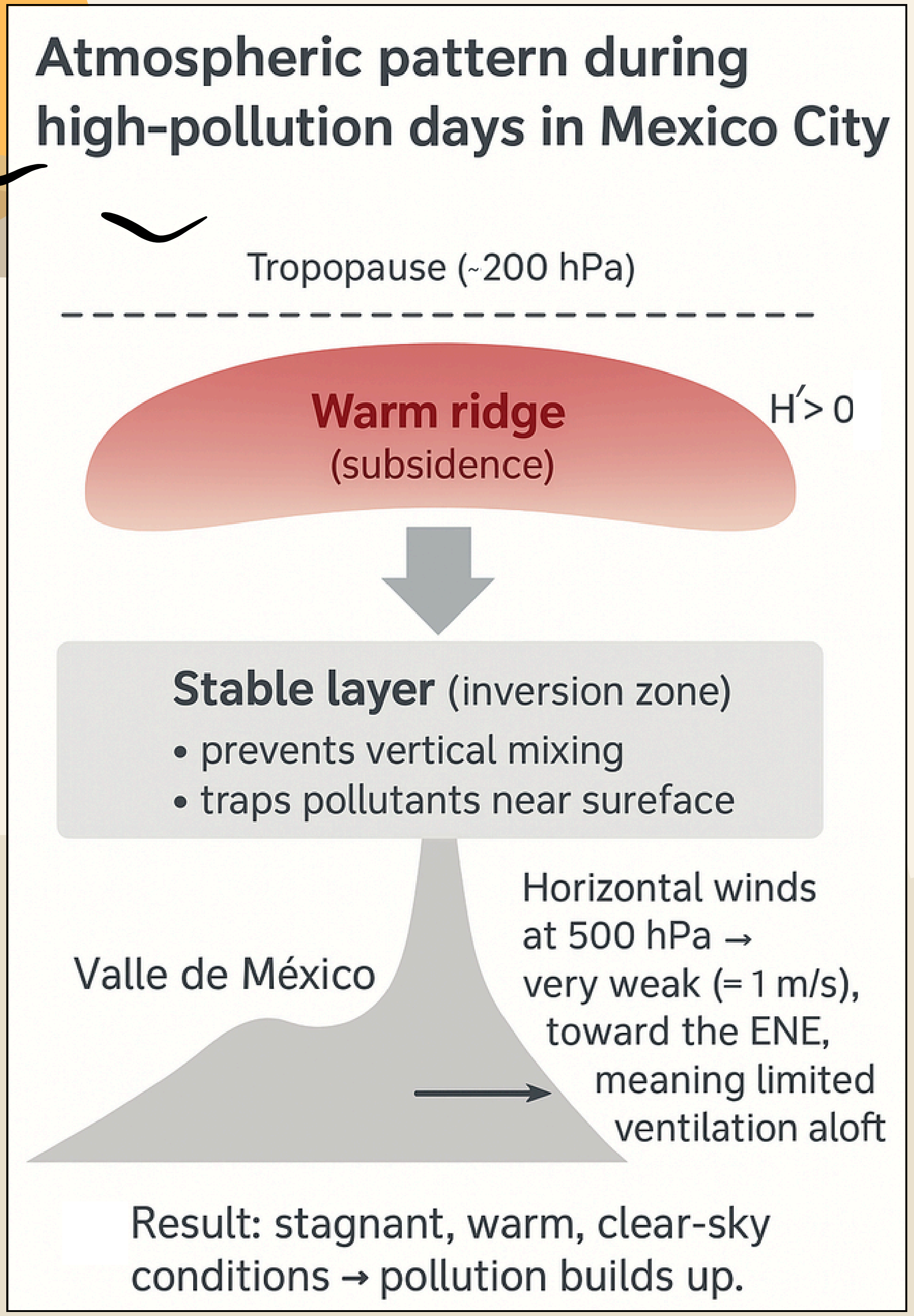


Fig. 10. Simplified schematic of the ridge-stagnation mechanism over the Mexico City basin: warm subsiding air aloft caps the cooler polluted layer below, limiting vertical mixing. (created by the author using Adobe Firefly).

- Diagnostic fingerprint: H′>0 ridge over central MX + weak ENE flow at 500 hPa.
- Subsidence warms mid-troposphere → stability
- Inversion cap forms → shallow mixing layer
- Weak winds → limited ventilation
- Convergence + topographic trapping

The ridge–stagnation regime is the primary synoptic driver of CDMX high-pollution days.

COMPARISON WITH HONG KONG

MX vs HK: Why High-Pollution Days Arise

- Local stagnation
- Ridge-driven subsidence
- Basin confinement
- Dry-season stability

VS

- Imported pollution
- Winter monsoon (northeasterlies)
- Tropical-cyclone subsidence
- Sea–land breeze recirculation
- Strong multi-pollutant coupling

Same mechanism (subsidence + weak winds), different origins.

SUMMING UP

CONCLUSIONS | LEARNING | NEXT STEPS

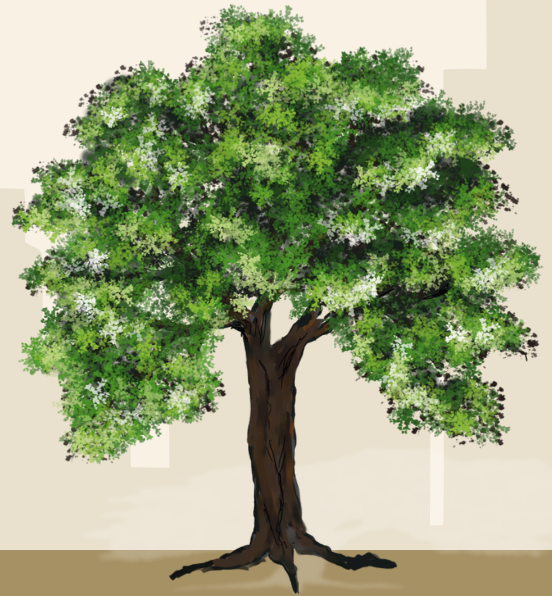
FINAL TAKEAWAYS

- Extreme pollution in CDMX = ridge-driven stagnation
- Mechanism is physically robust & consistent across pollutants
- HK = similar symptoms, different causes
- Composite meteorology → useful for alerts & forecasting
- Guided study fully achieved all objectives

LEARNING

& Accomplishments

- Built dataset from scratch (100+ Mexico City files)
- Learned specialized Python libraries (xarray, Cartopy)
- Developed full synoptic composite pipeline
- Understood ridge dynamics + ABL suppression
- Integrated cross-city literature



WHAT'S NEXT?

Toward a publishable extension of this study

- **Expand the synoptic framework beyond 500 hPa:**
  - Test 700-hPa composites instead of 850 hPa → more relevant for a high-altitude basin like MCMA.
  - Assess whether the ridge signature strengthens/weakens across layers → contributing to a vertical-structure analysis.
- **Develop a multi-component extreme-event classifier:**
  - Move beyond single-pollutant p_{90} thresholds.
  - Use clustering / PCA / percentile intersections to detect multi-pollutant extreme days.
  - Could reveal hidden "joint-exposure" patterns relevant for health-risk communication.
- **Construct a full synoptic composite atlas for Mexico City:**
  - Seasonal composites
  - ENSO-phase composites
  - Cold-front vs. non-front conditions
- **Build a parallel Hong Kong composites:**
  - Same pipeline, same spatial scale → first apples-to-apples comparison MX–HK.
  - It could explain structural differences in pollution resilience between a basin and a monsoon coastal city.
- **Integrate trajectories (HYSPLIT) for dust or transport analysis:**
  - Especially for O_3 and PM events.
  - Would deepen the "transport vs stagnation" distinction.
- **Define "ridge precursors" for early warning:**
  - Identify pressure & wind anomalies 1-2 days before events.
  - Practical impact for forecasting systems (IAyS communication and SEDEMA alt couldlert timing).
- **Formulate a publication pathway → MSc thesis plan:**
  - Paper I: MX synoptic controls (this report expanded)
  - Paper II: Multi-pollutant extreme event classification
  - Paper III: MX vs HK comparative synoptic climatology

THANK YOU

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